P-S-1792
July 10, 1973
SUPERSEDING
(see 6.4)

FEDERAL SPECIFICATION

SOAP, LAUNDRY (NEUTRAL AND BUILT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers four types of laundry soaps, with and without alkali, suitable for laundering with soft water (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The soaps covered by this specification shall be of the following types, classes, and quantities, as specified (see 6.2):

Type I - Neutral, high temperature (without added alkali).

Class 1 - Chip
Class 2 - Powdered

Type II - Neutral, low temperature (without added alkali).

Class 1 - Chip
Class 2 - Powdered

Type III - Built, high temperature (with added alkali), powdered.

Type IV - Built, low temperature (with added alkali), powdered.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-566 - Boxes, Folding, Paperboard.

FSC 7930

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PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-45 - Packaging Of Bulk Quantities Of Soaps, Detergents And Related Products (Chip, Powdered, Or Granular Form) For Domestic And Overseas Shipment.
PPP-P-1660 - Pallet, Expendable.
PPP-S-760 - Strapping, Nonmetallic (And Connectors).

Federal Standards:

FED-STD-123 - Marking For Domestic Shipment (Civil Agencies).
FED-STD-536 - Soap And Soap-Products (Including Synthetic Detergents); Sampling And Testing.
FED-STD-595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal ecifications and Standards, as outlined under General Information the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS:

21 CFR 191 - Federal Hazardous Substances Act.

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(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

American Society for Testing and Materials

ASTM D460-57T	-	Soap and Soap Products, Sampling and Chemical Analysis Of
ASTM D800-58	-	Industrial Metal Cleaning Compositions, Chemical Analysis Of
ASTM D820-58	-	Soaps Containing Synthetic Detergents, Chemical Analysis Of
ASTM D1172-56	-	pH of Aqueous Solutions of Soaps and Detergents, Test For

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Soap and Detergent Association

Test Procedure and Standards - ABS and LAS Biodegradability Scientific and Technical Report No. 3.

(Application for copies should be addressed to the Soap and Detergent Association, 475 Park Avenue South, New York, NY 10016.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distribut among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Composition. When tested as specified in 4.3, the composition of the laundry soaps shall conform to the requirements specified in table I.

3.2 Certificate of compliance. The supplier shall submit to the contracting officer a certificate of compliance certifying that the organic oxidation inhibitor content complies with the requirements in table I. The contractor shall state which inhibitor and the percent of inhibitor in the final product has been used, and shall not change inhibitor during production without approval of the contracting officer. The supplier shall also submit a certificate of compliance stating that no enzyme additives, derived from spore-forming bacteria, in excess of residual amounts found in raw materials, have been made a part of formulations provided under this specification.

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TABLE I. Composition

Characteristic	Type I Class 1 Min.	Type I Class 1 Max.	Type I Class 2 Min.	Type I Class 2 Max.	Type II (Class I, II) Min.	Type II (Class I, II) Max.	Type III Min.	Type III Max.	Type IV Min.	Type IV Max.
Moisture and volatile matter at 105°C, percent	-	10.0	-	6.0	-	7.0	-	6.0	-	16.0
Sum of free alkali or free acid, total matter insoluble in alcohol and sodium chloride, percent	-	6.0	-	6.0	-	8.0	-	-	-	-
Free alkali, calculated as sodium hydroxide, (NaOH), percent	-	0.2	-	0.2	-	0.2	-	-	-	0.2
Free acid, calculated as oleic acid, percent	-	0.2	-	0.2	-	0.2	-	-	-	-
Matter insoluble in water, percent	-	1.0	-	1.0	-	1.5	-	1.0	-	1.0
Anhydrous soap content, percent	85.0	-	89.0	-	86.0	-	29.0	-	50.0	-
Titer of the mixed fatty acids prepared from the soap, °C.	39.0	-	39.0	-	-	18.0	39.0	-	-	18.0
Water soluble sodium silicate, calculated as 1 Na_2O:3.25 SiO_2, percent	-	-	-	-	2.0	4.0	-	-	-	-
Copper, parts per million	-	-	-	-	-	10.0	-	-	-	-

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TABLE I. Composition (cont'd)

Characteristic	Type I Class 1 Min.	Type I Class 1 Max.	Type I Class 2 Min.	Type I Class 2 Max.	Type II (Class I, II) Min.	Type II (Class I, II) Max.	Type III Min.	Type III Max.	Type IV Min.	Type IV Max.
Iodine number (WiJS) of the mixed fatty acids prepared from the soap	-	-	-	-	80.0	95.0	-	-	-	-
Organic oxidation inhibitor, percent 1/	-	-	-	-	0.05	-	-	-	-	-
Unsaponified saponifiable matter, percent	-	-	-	-	-	1.0	-	-	-	-
Synthetic detergent (100% active), percent	-	-	-	-	-	-	2.0	-	-	-
Silicates (expressed as SiO_2), percent	-	-	-	-	-	-	15.0	-	-	-
Phosphates (expressed as P_2O_5), percent	-	-	-	-	-	-	2.8	20.0	-	-
Optical brighteners, percent 2/	-	-	-	-	-	-	0.13	-	-	-
Alkalinity (expressed as Na_2O), percent	-	-	-	-	-	-	32.0	-	20.0	-
Sodium carboxymethylcellulose (100% active), percent	-	-	-	-	-	-	0.45	-	-	-
Biodegradability of synthetic detergent, percent	-	-	-	-	-	-	90	-	-	-

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TABLE I. Composition (cont'd)

Characteristic	Type I Class 1 Min.	Type I Class 1 Max.	Type I Class 2 Min.	Type I Class 2 Max.	Type II (Class I, II) Min.	Type II (Class I, II) Max.	Type III Min.	Type III Max.	Type IV Min.	Type IV Max.
pH of 1% solution, units	-	-	-	-	-	-	11.5	12.0	11.0	11.5

1/ Acceptable inhibitors of the following types shall be used unless otherwise specified: diphenylamine; orthodiphenylbiguanidine; paratertiaryamylphenol; paratertiarybutylphenol. Supplier shall indicate which inhibitor will be used, and shall not change inhibitor during production unless approved by the contracting officer (see 3.2).

2/ Optical brightener shal be Geigy Tinopal AMS or equal.

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3.3 Color. The soaps (type I and II) shall have a uniform, light color. The color of a freshly sampled soap shall be not darker than color number 30266 of FED-STD-595 (see 6.3). The soaps (types III and IV) shall also have a uniform, light color and shall be no darker than color number 33434 of FED-STD-595 (see 6.3).

3.4 Odor. The soaps shall be free from a rancid, fishy, or horsey odor. The product shall not develop a rancid, fishy, or horsey odor when tested as specified in 4.3.

3.5 Form.

3.5.1 Types I and II, class 1. Types I and II, class 1 soaps shall be in chip or flake form. No particle shall exceed 3 grams in weight when tested as specified in 4.3.

3.5.2 Types I and II, class 2 and types III and IV. Types I and II, class 2, and types III and IV soaps shall be in powder or granular form. Not more than 2 percent shall be retained on a number 12 sieve and not more than 20 percent shall pass through a number 140 sieve when tested as specified in 4.3.

3.6 Solubility.

3.6.1 Types I and III. Types I and III soaps shall be readily soluble in water at 140°F when tested as specified in 4.3.

3.6.2 Types II and IV. Types II and IV soaps shall be readily soluble in water at 100°F when tested as specified in 4.3.

3.7 Labeling. Each unit container of laundry soap shall be durably and legibly marked with the following information and, in addition, types III and IV shall be labeled in conformance with the Federal Hazardous Substances Act.

3.7.1 Type I. "SOAP LAUNDRY, -NEUTRAL, HIGH TEMPERATURE
(without added alkali)
P-S-1792
TYPE I"
(CLASS)
(FEDERAL STOCK NUMBER)
(MANUFACTURER'S NAME AND ADDRESS)
(DATE OF PACK AND LOT NUMBER)

"A neutral laundry soap, free from alkali, suitable for use in high temperature (140° to 160°F.) laundering of cotton fabrics. Do not use on wool, silk, or synthetic fabrics.

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3.7.2 Type II. "SOAP LAUNDRY - NEUTRAL, LOW TEMPERATURE
(without added alkali)
P-S-1792
TYPE II"
(CLASS)
(FEDERAL STOCK NUMBER)
(MANUFACTURER'S NAME AND ADDRESS)
(DATE OF PACK AND LOT NUMBER)

"A neutral laundry soap, free from alkali, suitable for use in low temperature (90° to 100°F.) laundering of cotton, silk, wool, and synthetic fabrics and finished surfaces.

3.7.3 Type III. "SOAP LAUNDRY - BUILT, HIGH TEMPERATURE
(with added alkali)
P-S-1792
TYPE III"
(FEDERAL STOCK NUMBER)
(MANUFACTURER'S NAME AND ADDRESS)
(DATE OF PACK AND LOT NUMBER)

"A built laundry soap, with an alkali, suitable for use in high temperature (140° to 160°F.) laundering of cotton fabrics. Do not use on silk, woolen, or synthetic fabrics.

CAUTION: ALKALINE POWDER
KEEP OUT OF REACH OF CHILDREN.
AVOID CONTACT WITH SKIN AND EYES.
IN CASE OF CONTACT, IMMEDIATELY FLUSH SKIN OR EYES WITH PLENTY OF WATER.
FOR EYE CONTACT, CONTINUE FLUSHING WITH WATER FOR AT LEAST 15 MINUTES AND GET MEDICAL ATTENTION IMMEDIATELY.
AVOID BREATHING DUST FROM POWDER.
DO NOT TAKE INTERNALLY.
DO NOT MIX WITH ACID OR OTHER CHEMICALS.

3.7.4 Type IV. "SOAP LAUNDRY - BUILT, LOW TEMPERATURE
(with added alkali)
P-S-1792
TYPE IV"
(FEDERAL STOCK NUMBER)
(MANUFACTURER'S NAME AND ADDRESS)
(DATE OF PACK AND LOT NUMBER)

"A built laundry soap, with an alkali, suitable for use in low temperature (90° to 100°F.) laundering of cotton fabrics. Do not use on silk, woolen, or synthetic fabrics.

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CAUTION: ALKALINE POWDER
KEEP OUT OF REACH OF CHILDREN.
AVOID CONTACT WITH SKIN AND EYES.
IN CASE OF CONTACT, IMMEDIATELY FLUSH SKIN OR EYES WITH PLENTY OF WATER.
FOR EYE CONTACT, CONTINUE FLUSHING WITH WATER FOR AT LEAST 15 MINUTES AND GET MEDICAL ATTENTION IMMEDIATELY.
AVOID BREATHING DUST FROM POWDER.
DO NOT TAKE INTERNALLY.
DO NOT MIX WITH ACID OR OTHER CHEMICALS.

3.8 Workmanship. The finished product shall be clean, uniform, free flowing, and shall not lump or cake.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for examination and tests shall consist of all the end item of one type and class prepared in a single batch and submitted for inspection at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards, unless otherwise excluded, amended or qualified in this specificaiton or applicable purchase documents.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.4. Samples shall be

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drawn from each lot of end items offered for inspection. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of one filled primary container of the specified capacity for examination in 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3, and in units of shipping containers for examination in 4.2.3.

4.2.2.1.1 Examination of the primary container for visual defects in construction and markings. The sample unit for this examination shall be one filled primary container.

Examine	Defect
Construction	Not in container specified. Any leakage or sifting of contents. Any split, break, dent, hole, tear, or puncture. Seams or closures not as specified for liner or container.
Markings (label or printing)	Omitted, illegible, incorrect, incomplete, not as specified. Label or printing not clear, smeared.

4.2.2.1.2 Examination for defects in net contents. The sample unit for this examination shall be one filled primary container. The lot shall be unacceptable if the average net contents per container is less than the specified or indicated quantity.

4.2.2.1.3 Examination of the end item for defects in workmanship, color and odor. The sample unit for this examination shall be the contents of one filled primary container.

Examine	Defect
Workmanship	Not type specified (chip or powdered). Not clean; not uniform; not homogeneous in appearance; any foreign matter. Not free flowing; any lumping or caking.
Color	Not uniform or as specified. Darker than color chip Number 30266 (types I and II) or darker than color Number 33434 (types III and IV).
Odor	Any rancid, fishy, or horsey odor.

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4.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping container in the end item inspection lot.

Examine	Defect
Marking (exterior and interior, when applicable)	Omitted, incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: fiberboard container or interior package (when applicable) not as specified; incomplete closure of container flaps inadequate stapling, improper taping or loose strapping: bulged or distorted container.
Content	Number of interior packages (when applicable) per shipping container is more or less than specified.
Weight	Quantity of laundry soap is more or less than that indicated by the specified interior package.

4.2.4 Inspection levels and AQLs for examinations. The inspection levels, for determining the sample size, and the AQLs, expressed in defects per hundred units, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.2.2.1.1	S-4	2.5
4.2.2.1.2	S-2	N/A
4.2.2.1.3	S-2	2.5
4.2.3	S-2	4.0

4.3 Testing of the end item. The soap shall be tested for the characteristics specified in table I in accordance with test methods specified in table II for each lot presented for examination. The lot size shall be expressed in

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units of 1 pound each of the laundry soap. The sample unit, shall be a 1 pound composite sample selected in accordance with 4.3.1. The sample size shall be as shown in the table below. Two specimens of each sample unit, for moisture and volatile matter determinations, shall be weighed immediately upon opening the sealed container. Care shall be exercised to prevent the loss or absorption of moisture or loss of volatile matter during the sampling, weighing and storage of the specimens and samples. The lot shall be unacceptable if one or more sample units or the composite sample fails to meet any test requirement specified.

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II. Instructions for testing

Characteristic	Applicable to types	Specification reference		Number Determinations on composite sample	Results reported as	
		Requirement	Test method		Pass or fail	Numerically to nearest
Moisture and matter volatile at 105°C.	I,II,III,IV	Table I	201.1*	Average of 2		0.1 percent
Anhydrous soap content	I,II,IV	Table I	401.3	" " "		"
Anhydrous soap content	III	Table I	5601.1*	" " "		"
Matter insoluble in water	I,II,IV	Table I	303.1*	" " "		"
Matter insoluble in water	III	Table I	5801*	" " "		"
Free alkali or free acid	I,II,IV	Table I	302.3*	" " "	"	"
Alcohol-insoluble matter	I,II	Table I	301.3*	" " "		"
Chlorides, as sodium chloride	I,II	Table I	501*	" " "		"
Unsaponified saponifiable matter	II	Table I	603.1*	" " "		"
Water-soluble sodium silicate as Na_2O: 3.25 SiO_2	II	Table I	1301*	" " "		"
Iodine number (WiJS) of mixed fatty acids	II	Table I	1101.2*	" " "		Number
Titer of mixed fatty acids	I,II,IV	Table I	901* 5601.1 3/	" " "		0.1 degree
Titer of mixed fatty acids	III	Table I	901*	" " "		"

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TABLE II. Instructions for testing (cont'd)

Characteristic	Applicable to types	Specification reference: Requirement	Specification reference: Test method	Number Determinations on composite sample	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Copper content	II	Table I	ASTM D460-57T	Average of 2		ppm
Organic oxidation inhibitor content	II	Table I	2/	2/		0.01 percent
Total alkalinity as Na_2O	III,IV	Table I	ASTM D800	Average of 2		0.1 percent
pH of 1 percent solution	III,IV	Table I	ASTM D1172-56	" " "		0.1 unit
Synthetic detergent content (chloroform solubles)	III	Table I	4.4.4	" " "		0.1 percent
Silicate content as SiO_2	III	Table I	6001*	" " "		"
Phosphate content as P_2O_5	III	Table I	ASTM D460-57T	" " "		"
Sodium carboxymethylcellulose content	III	Table I	ASTM D820-58	" " "		"
Optical brightener content	III	Table I	ANAL ABS 14, #2069	" " "		0.01 percent
Color	I,II,III,IV	3.3	FED-STD-595	1	X	-
Odor and stability	I,II,III,IV	3.4	4.4.2	1	X	-

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TABLE II. Instructions for testing (cont'd)

Characteristic	Applicable to types	Specification reference		Number Determinations on composite sample	Results reported as	
		Requirement	Test method		Pass or fail	Numerically to nearest
Form (particle weight)	I,II (class 1)	3.5.1	4.4.1	1	X	-
Form (particle size)	I,II(class 2) III,IV	3.5.2	2101*	Average of 2		1 percent
Solubility	I,II,III,IV	3.6	4.4.3	-	X	-
Biodegradability of synthetic detergent 4/	III	Table I	4.4.5	Average of 2		0.1 percent

1/ All tests marked with * are performed in accordance with FED-STD-536.

2/ Certificate of compliance (see 3.2).

3/ The mixed fatty acids shall be prepared in accordance with procedure outlined in 3.1, 3.2, 3.3 and 3.4 of Test Method 5601.1.

4/ The testing shall be performed, and actual test data (field or laboratory) furnished on the first lot only. For each subsequent lot, a certificate shall be furnished, indicating compliance with this requirement. If at any time there is a change in formulation, testing again shall be performed and data submitted as for the first lot.

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4.3.1 Composite sample. When four or more containers comprise the shipping lot, a sample of not less than 1 pound shall be taken from each of not less than four separate containers, selected throughout the lot and presented for examination. When less than four shipping containers comprise the lot, the total samples shall consist of not less than three sample portions taken at random from the containers or container. The samples shall be placed in a receptacle and mixed. A 1-pound composite sample shall be placed in an air-tight container, sealed, and marked with the identifying lot number and date.

4.4 Test procedures.

4.4.1 Particle weight, types I, II (class 1). From a 100-gram sample taken from the composite sample, select 10 of the largest observed particles. Each particle shall be weighed separately. The weight of each particle shall not exceed 3 grams.

4.4.2 Odor and stability test (type II). A quart size glass fruit jar shall be half filled with the product. A 15 by 175 millimeter test tube, two-thirds full of water, shall be placed in the center of the soap product so that the bottom of the test tube rests on the bottom of the jar and so that the water vapor may freely escape into the jar. The soap product shall be lightly packed by gently tapping on the table top. Soap shall be added during tapping so that the jar remains half filled. The jar shall be sealed with a gasket, making sure that no metal is exposed within the jar. The jar shall be placed in an oven or cabinet and maintained at 125° to 130°F. for a period of 15 days. At the end of 15 days, the cover shall be removed and the odor determined and compared with the orginal soap as received.

4.4.3 Solubility in water (types III and IV). Add 0.5 grams of soap to 100 grams of water 100° ± 2°F. or 140° ± 2°F., as applicable. After moderate stirring for 10 minutes, the soap solution shall be clear and free from sediment.

4.4.4 Synthetic detergent (chloroform-soluble matter) (type III). Eight (8 ± 0.001) g of type III soap shall be weighed and mixed with about 16 g of clean dry sand (20-30 mesh Ottowa sand) and placed into a 25 by 80 mm extractor thimble (Whatman). A thin layer of sand shall be placed at the top of the soap-sand mixture. The thimble and contents shall be placed in a Soxhlet extraction apparatus and extracted with chloroform, U.S.P. grade, for 16 hours. The chloroform extract shall be transferred quantitatively to a tared beaker and the chloroform evaporated on a steam bath until the odor of chloroform disappears. Dry the beaker and chloroform-soluble matter in oven (103 ± 2°C) for 30 minutes, cool in dessicator and weigh. Calculate percent chloroform-soluble matter.

$$\text{Percent chloroform-soluble matter} = \frac{\text{Weight of residue} \times 100}{\text{Weight of soap sample}}$$

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4.4.5 Biodegradability of synthetic detergent (type III).

4.4.5.1 Anionic synthetic detergents. The supplier shall submit a certificate of compliance with the requirements for biodegradability. The certificate shall be accompanied by actual test data (field or laboratory), including the test procedure utilized in making the biodegradability determination. When the detergent is an alkyl benzene sulfonate or a linear alkylate sulfonate, the biodegradability shall be determined in accordance with the Test Procedure and Standards ABS and LAS Biodegradability Scientific and Technical Report No. 3 of the Soap and Detergent Association.

4.4.5.2 Nonionic and other synthetic detergents. Until an acceptable method of determining the biodegradability of a nonionic detergent is agreed upon by industry and is acceptable to the Government, the supplier shall certify that his product meets the 90 percent biodegradability requirement when tested in accordance with one of the current test methods. The certificate shall be accompanied by actual test data (field or laboratory), indicating test method used in making the biodegradability determination.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-pound quantities. One-pound quantities of laundry soap shall packaged in a snug-fitting siftproof folding paperboard box conforming to variety 2, style I, type A, class a or b of PPP-B-566. Each box shall be securely closed in accordance with the applicable requirements specified in the appendix of the box specification.

5.1.1.2 Fifty-pound, 100-pound, and 200-pound quantities. Fifty-pound, 100-pound or 200-pound quantities of laundry soap shall not require packaging.

5.1.2 Level B (civil agencies). One-pound of laundry soap shall be packaged in a box normally used for the item. The box shall be secured to prevent sifting of contents. When used, liners or wraps shall be of suitable material which shall not affect nor be adversely affected by the laundry soap.

5.1.3 Level C. Laundry soap shall be packaged to afford adequate protection against physical damage and deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard paractice when it meets these requirements.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-pound quantities. Forty-eight 1-pound quantities of laundry soap of one description only, packaged as specified in 5.1.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed method V, and reinforced as specified in the appendix of the container specification.

5.2.1.2 Fifty-pound, 100-pound, and 200-pound quantities. Fifty-pound, 100-pound, or 200-pound quantities of laundry soap shall be packed in accordance with the applicable level A requirements specified in PPP-P-45.

5.2.2 Level B.

5.2.2.1 One-pound quantities (military requirements). Forty-eight 1-pound quantities of laundry soap of one description only, packaged as specified in 5.1.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method III as specified in the appendix of the container specification.

5.2.2.1.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.2.2 One-pound quantities (Civil Agencies). Forty-eight 1-pound quantities of laundry soap of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with the appendix of the container specification.

5.2.2.3 Fifty-pound, 100-pound, and 200-pound quantities. Fifty-pound, 100-pound or 200-pound quantities of laundry soap shall be packed in accordance with the applicable level B requirements specified in PPP-P-45.

5.2.3 Level C. Laundry soap, packaged as specified in 5.1 (when applicable), shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classificaiton Rules, as applicable.

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5.3 Unit loads. When specified (see 6.2), sacks and drums of soaps in the quantity specified shall be placed on a pallet conforming to PPP-P-1660, type II; style optional; class B or C; grade 2; size optional. For drums, edge protectors shall be placed on all four top edges of the pallet load and shall be of sufficient length to contact the outside edges of the corner containers on the pallet. The edge protectors shall be a strip of solid fiberboard a minimum of 6 inches wide and not less than 0.06 inch thick, with a score line running lengthwise; or laminated paperboard a minimum of 6 inches wide and not less than 0.225 inch thick, formed into a rigid angle. The legs of the edge protectors shall be a minimum of 2 inches on the side of the pallet load and 3 inches on the top of the load. For sacks, edge protectors shall be placed under the straps. The load shall be secured with steel or nonmetallic strapping conforming to QQ-S-781 or PPP-S-760. When there is only one tier of containers on the pallet, straps shall be placed girthwise around the load midway of the length of the containers. When there are two or three tiers of containers on the pallet, straps shall be placed girthwise around the load midway of the length of the containers in each tier, except sacks do not require these. A minimum of two straps shall be placed around the top, bottom and sides of the load and two straps around the top, bottom and ends of the loads. When steel strapping is applied to a load at right angles to each other and in the same plane, it shall be secured from lateral movement and displacement by the application of steel crossties of a size appropriate for the strapping used. Alternatively, the load shall be secured to the pallet by the shrink film process. Thermoplastic films of 6 mil polyethylene, 5.5 mil. reinforced polyethylene, or 2.5 mil. polyvinyl chloride shall be used. Shrinking of the film shall be in accordance with industry practice.

5.4 Marking.

5.4.1 Civil Agencies. In addition to any special marking required by the contract or order, interior packages (when applicable) and shipping containers shall be marked in accordance with FED-STD-123 and the applicable general marking requirements of PPP-P-45.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages (when applicable) and shipping containers shall be marked in accordance with MIL-STD-129 and the applicable general marking requirements of PPP-P-45.

5.4.3 Special marking. Each interior package (when applicable) and shipping container shall bear the date of manufacture of the laundry soap. In addition, each shipping container shall be printed or stenciled with the following legend: "STORE IN A COOL DRY PLACE".

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6. NOTES

6.1 Intended use. Type I, classes 1 and 2 neutral laundry soap, without an added alkali, is intended for high temperature laundering (140° to 160°F) of cotton fabrics. Type II, classes 1 and 2 neutral laundry soap, without added alkali, is intended for low temperature laundering of cotton, silk, woolen or synthetic fibers. Type III, built laundry soap, with an added alkali, is intended for high temperature laundering (140° to 160°F) of cotton fabrics. Type IV, built laundry soap, with an added alkali, is intended for low temperature laundering (90° to 100°F) of cotton fabrics. Types I, III, and IV soaps are not suitable for use on silk, woolen, or synthetic fibers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Types, classes, and quantities required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When level B packaging is required for Civil Agencies (see 5.1.2).
(e) When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1.1).
(f) When unit loads are required (see 5.3).

6.3 Color chips are available from all General Services Administration Business Service Centers.

6.4 Supersession data. This specification includes the requirement of P-S-579A, dated March 6, 1963; P-S-585A, dated December 12, 1962; P-S-00585B (GSA-FSS), dated June 5, 1970; and P-S-600C, dated December 12, 1962. The following is a classification comparison between the superseded specifications and this specification:

P-S-1792	P-S-579A	P-S-585A and P-S-00585B	P-S-600C
Type I, Class 1	Type I		
Class 2	Type II		
Type II, Class 1			Form 1
Class 2			Form 2
Type III		Type II	
Type IV		Type I	

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P-S-1792

Custodians:

Army - GL
Air Force - 84

User activity:

Navy - YD, SH

Preparing activity:

Army - GL

Civil Agency Interest:

GSA-FSS
VA-DMS

Project No. 7930-0272

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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